Some possibly useful equations for Exam 2

Capacitors: $q = CV$ Energy Stored: $U_C = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$

Parallel Plate: $C = \frac{\varepsilon_0 A}{d}$ with dielectric: $C = \frac{\kappa\varepsilon_0 A}{d}$ Spherical: $C = \frac{4\pi\varepsilon_0 R_1 R_2}{(R_2 - R_1)}$

Capacitors in parallel: $C_{eq} = C_1 + C_2 + C_3 + C_4 + ...$ All have the same voltage, V

Capacitors in series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} + ...$ All have the same charge, q

RC circuit charging source voltage, *V*: $q = CV\left(1 - e^{-t/RC}\right)$ $V_C = \frac{q}{C} = V\left(1 - e^{-t/RC}\right)$ $i = \frac{V}{R}e^{-t/RC}$

RC circuit discharging: $q = q_0 e^{-t/RC}$ $V_C = \frac{q_0}{C}e^{-t/RC}$ $i = -\frac{q_0}{RC}e^{-t/RC}$

Resistors in series: $R_{eq} = R_1 + R_2 + R_3 + R_4 + ...$ All have the same current, i

Resistors in parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} + ...$ All have the same voltage, V

Ohm's Law: $V = IR$ **Power:** $P = iV = i^2R = \frac{V^2}{R} = \frac{work}{time}$ **Energy:** $\Delta U = q\Delta V$ $\Delta E = \Delta U + \Delta K$

Loop Rule: The algebraic sum of changes in electric potential (voltage) around any closed loop in a circuit must be zero. (***Kirchhoff's Voltage Law – KVL***)

Resistance rule: The potential change across a resistor is equal to $-iR$ when going in the direction of conventional current, and $+iR$ when going against the current.

EMF rule: The potential change through an ideal *emf* device is $+E$ moving through it in the direction of the *emf* arrow (from – to +) and $-E$ moving through it in the opposite direction.

Junction rule: The sum of the currents entering any junction must be equal to the sum of the currents leaving the junction. (***Kirchhoff's Current Law – KCL***) Charge is conserved.

Temperature change of resistance: $R = R_0(1 + \alpha\Delta T)$